# ML + EKF for Battery State of Charge Estimation

## Evaluation report

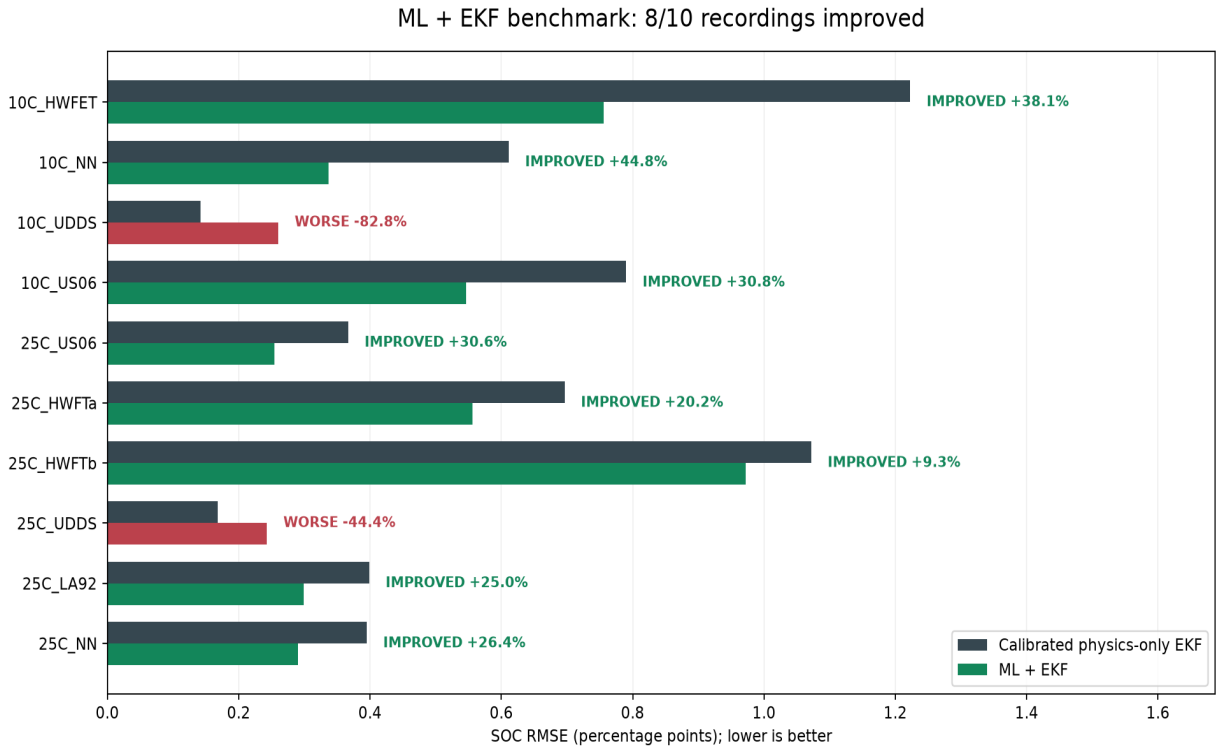
The trained feedforward network assists a calibrated extended Kalman filter. This report evaluates both estimators on ten Panasonic recordings using the same measurements and reference.

**Results:** 8 of 10 recordings improved. Mean recording RMSE: EKF 0.5863 pp versus ML + EKF 0.4516 pp, relative reduction 23.0%.

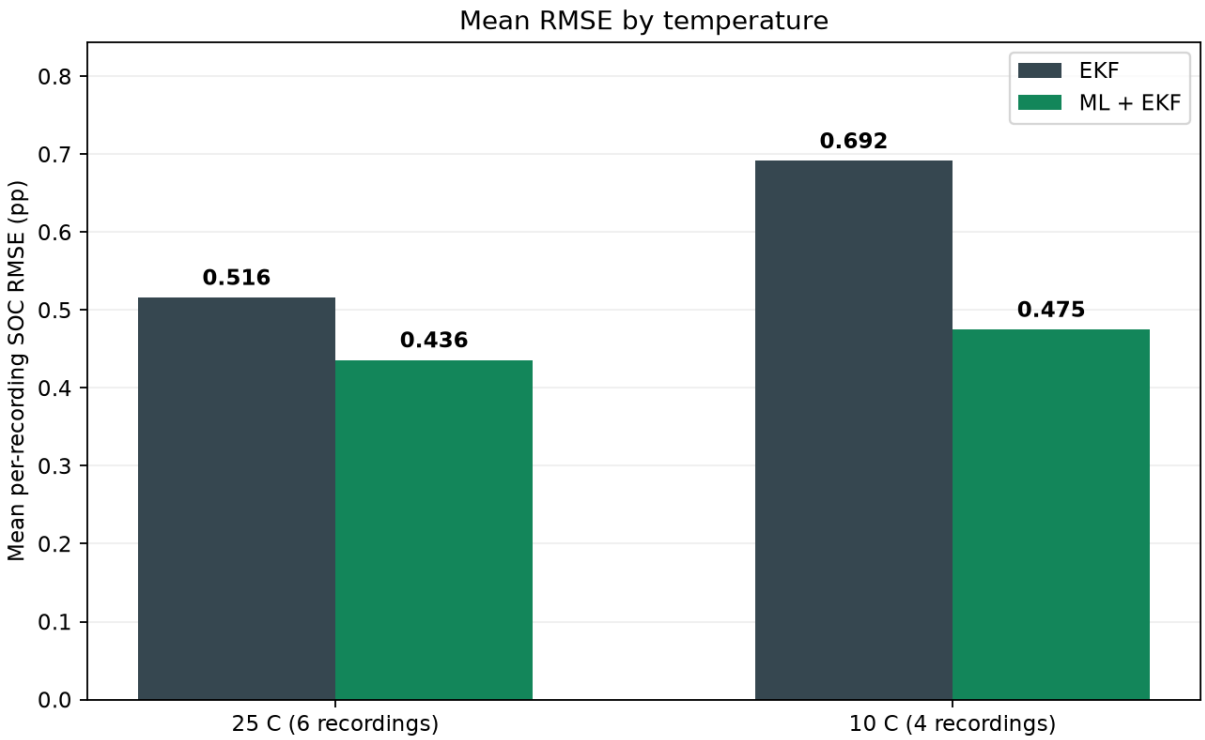
| Recording | EKF RMSE | ML + EKF RMSE | Gain vs EKF |
|---|---|---|---|
| 10C_HWFET | 1.223 pp | 0.756 pp | +38.1% |
| 10C_NN | 0.611 pp | 0.337 pp | +44.8% |
| 10C_UDDS | 0.142 pp | 0.260 pp | -82.8% |
| 10C_US06 | 0.790 pp | 0.547 pp | +30.8% |
| 25C_US06 | 0.367 pp | 0.255 pp | +30.6% |
| 25C_HWFTa | 0.696 pp | 0.556 pp | +20.2% |
| 25C_HWFTb | 1.072 pp | 0.973 pp | +9.3% |
| 25C_UDDS | 0.168 pp | 0.243 pp | -44.4% |
| 25C_LA92 | 0.399 pp | 0.299 pp | +25.0% |
| 25C_NN | 0.395 pp | 0.291 pp | +26.4% |

The two UDDS cases become worse. Results come from one cell, previously examined recordings and an Ah-derived reference using assumed 2.79826 Ah; improvements are not independent cross-cell proof.

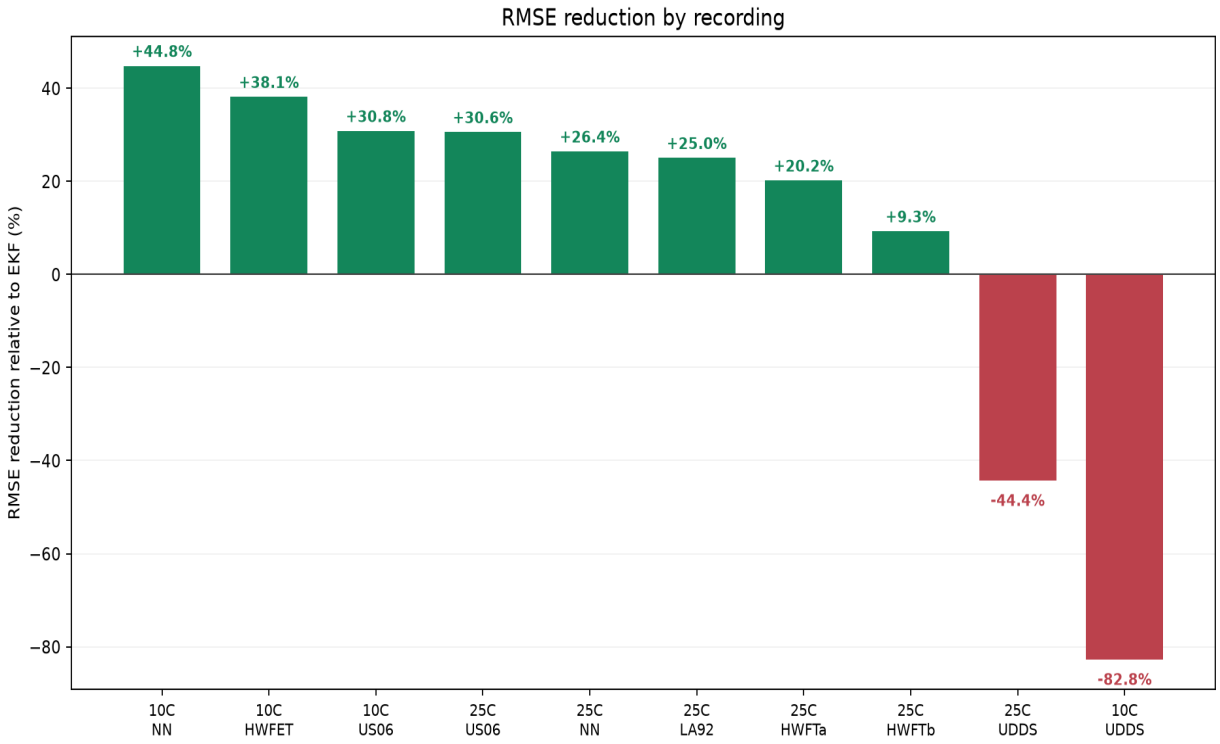
## Per-recording performance



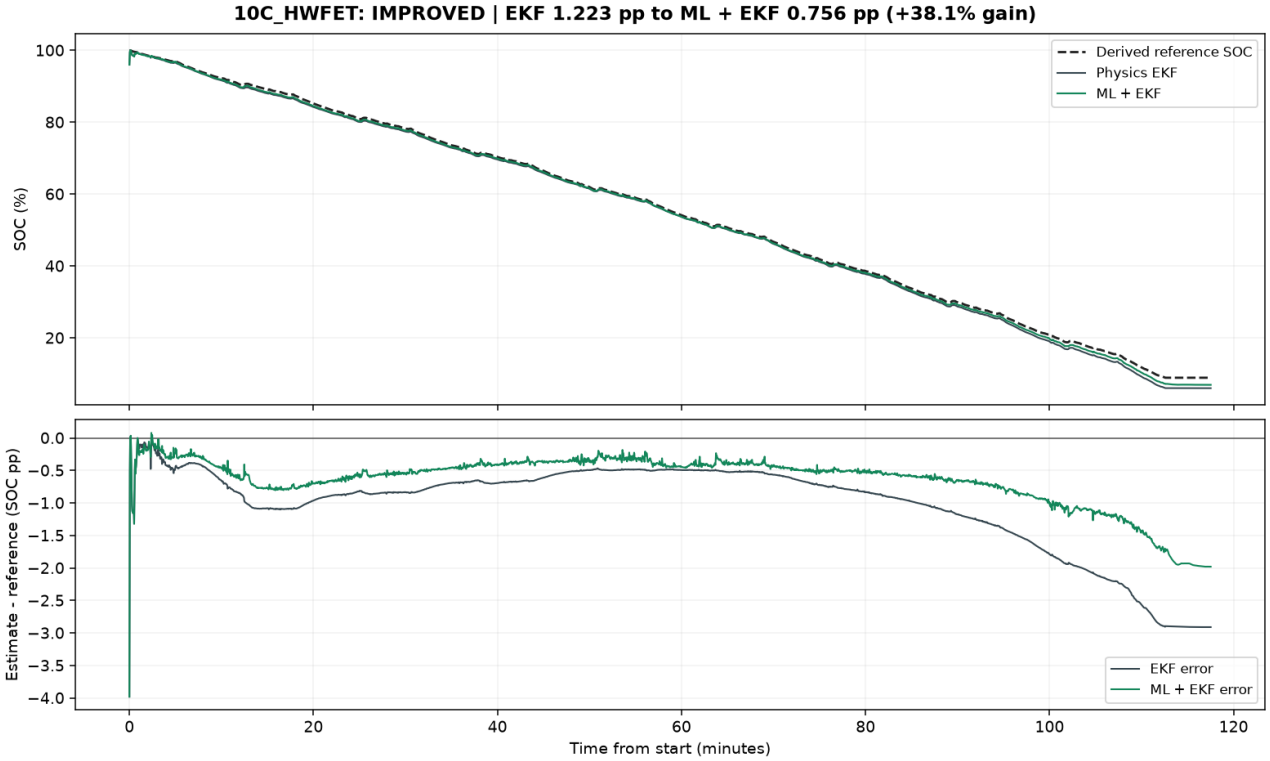
## Results by temperature



# Eight successes and two failures

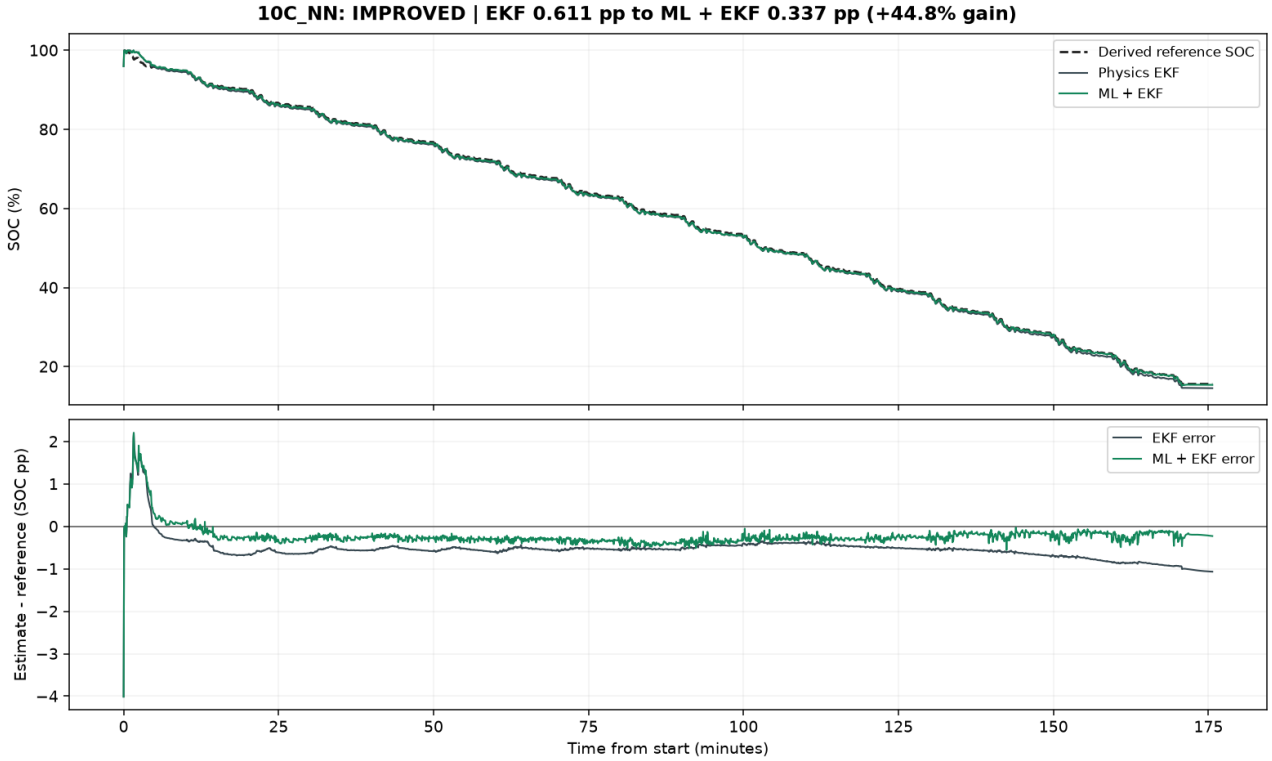


# Highway driving / cooler battery - useful reduction



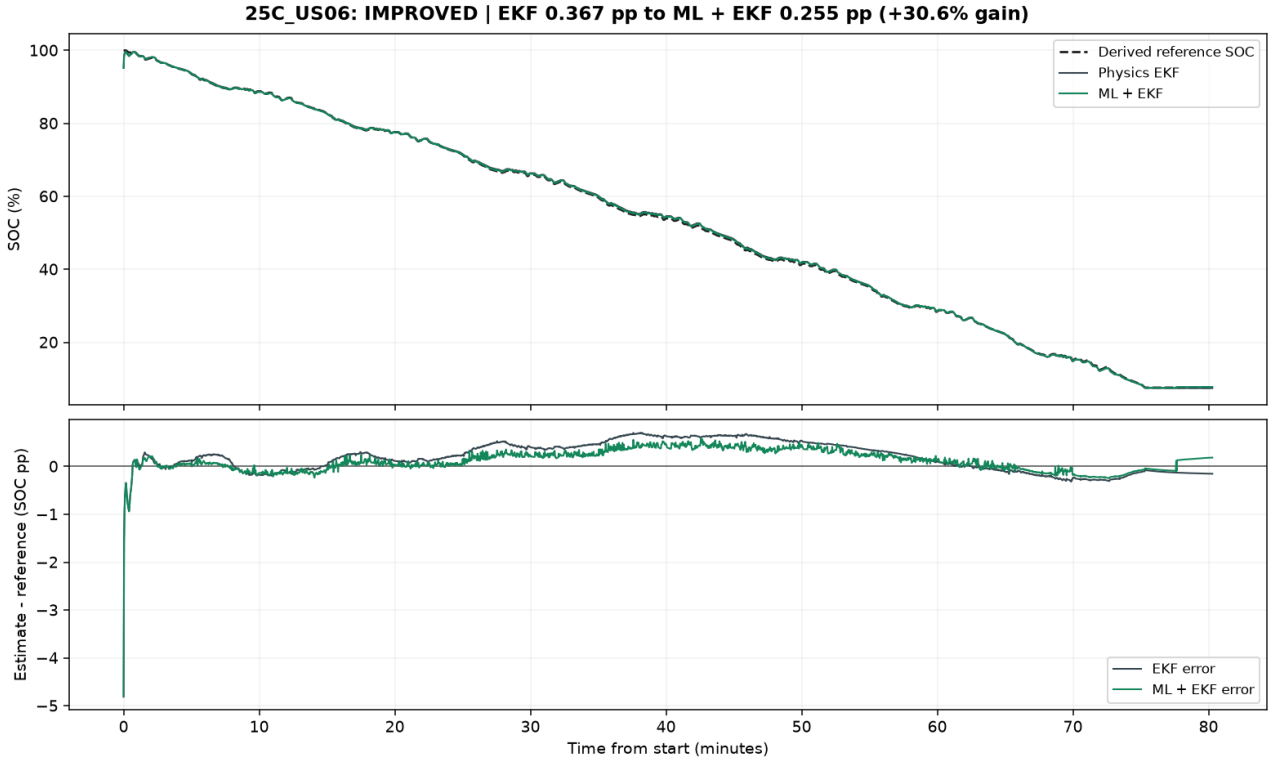

Top: reference and both SOC estimates. Bottom: signed SOC estimation error. Reference SOC is constructed from tester Ah counter and assumed capacity. Real-world benefits such as better remaining-range estimation are potential applications, not demonstrated on a vehicle.

## Mixed driving - largest relative improvement



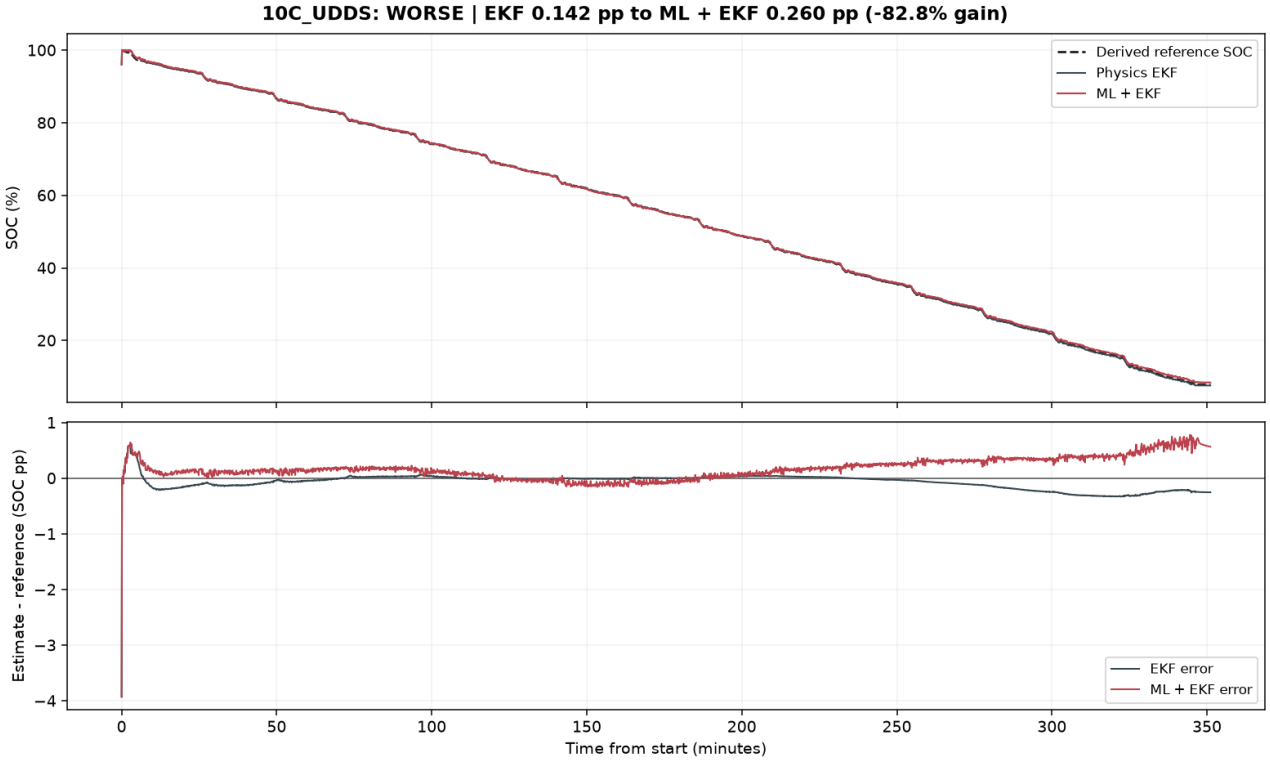

Top: reference and both SOC estimates. Bottom: signed SOC estimation error. Reference SOC is constructed from tester Ah counter and assumed capacity. Real-world benefits such as better remaining-range estimation are potential applications, not demonstrated on a vehicle.

# Aggressive acceleration - improved




Top: reference and both SOC estimates. Bottom: signed SOC estimation error. Reference SOC is constructed from tester Ah counter and assumed capacity. Real-world benefits such as better remaining-range estimation are potential applications, not demonstrated on a vehicle.

# City-driving UDDS - correction unnecessarily worsens accuracy




Top: reference and both SOC estimates. Bottom: signed SOC estimation error. Reference SOC is constructed from tester Ah counter and assumed capacity. Real-world benefits such as better remaining-range estimation are potential applications, not demonstrated on a vehicle.